Robot Perception, Localization and Navigation (Neoscholar)

Robot Operating System (ROS) Setup

The goal of this tutorial is to set up the Robot Operating System (ROS) on your computer, so you can learn and practise ROS.

Installation of Required Software

1. Download and install Docker Desktop: https://docs.docker.com/desktop
2. Download and install Visual Studio Code: https://code.visualstudio.com/download

ROS Docker Image

Download files

1. Download the ROS Docker image fine named `ROS-Docker.zip`, and unzip it into the folder you would like to work on for this module.
2. Enter the following command under the unzipped folder:

```
$ cd neoscholar-ros/simulator-robot-docker
$ docker build -f Dockerfile -t devrt/simulator-robot:latest .
```

You may need to wait for a while to have this docker image built. Once this is finished, move on to next step.

3. Enter the following command under the the unzipped folder:

```
$ cd neoscholar-ros/neoscholar-docker-image
$ docker compose up
```

It will take a while to install and set up the ROS docker image.

Use this ROS dev container with VSCode

1. Install Remote Development" extension on your VSCode.
2. Click `View` -> `Command Palette` -> `Dev Containers: Open Folder in Container` -> Open the `neoscholar-docker-image` folder you downloaded previously.
3. You need to wait a while for container to come up (only required once).

For detailed instructions, see: https://code.visualstudio.com/docs/remote/containers

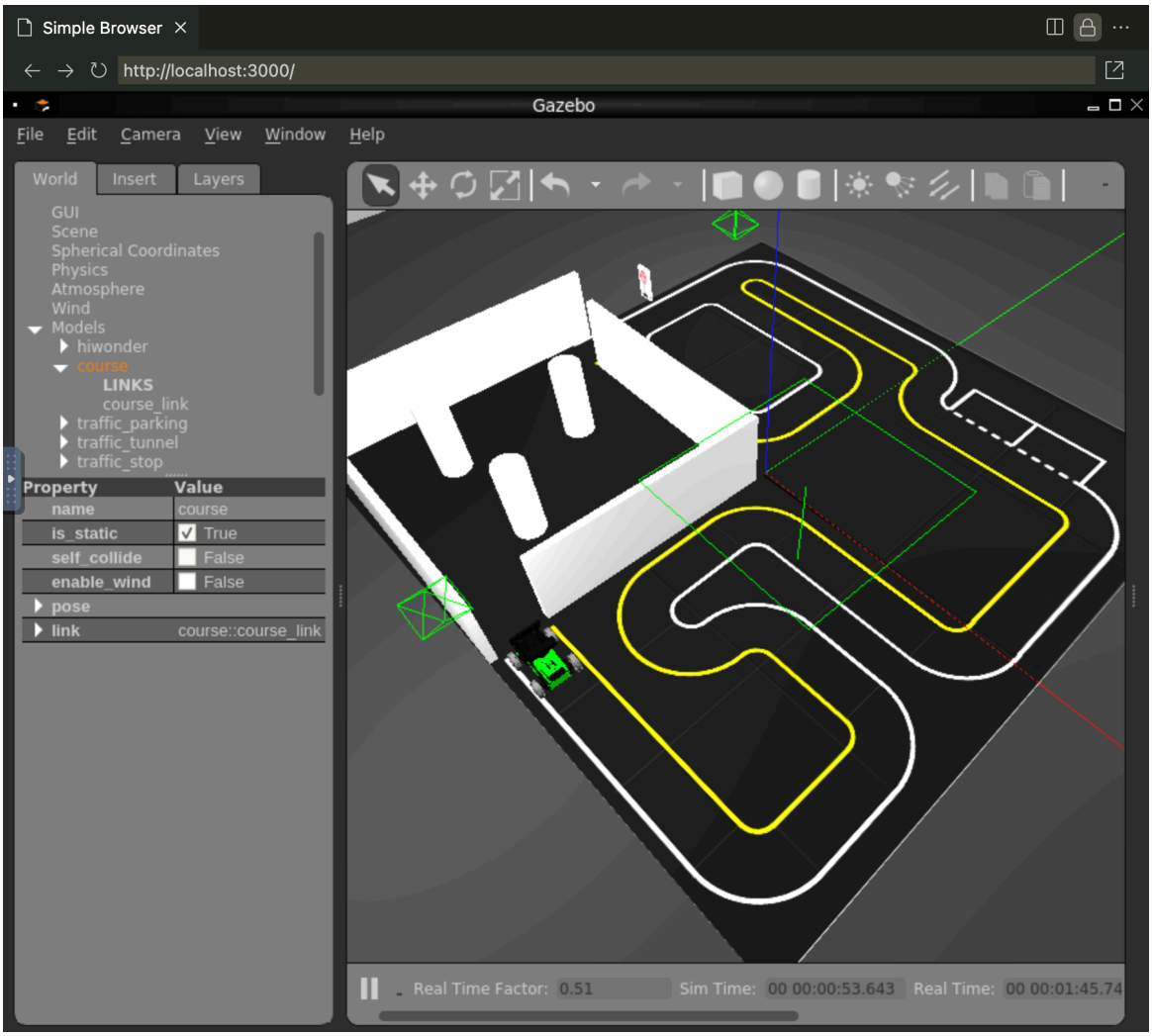
Now you should be in the ROS container where you can learn and practise ROS.

Open X11 server screen

1. Wait for the container to start.
2. Open http://localhost:3000/ using your favourite browser.

If you want browser screen to be integrated with VSCode, use Browser Preview for VS Code extension.

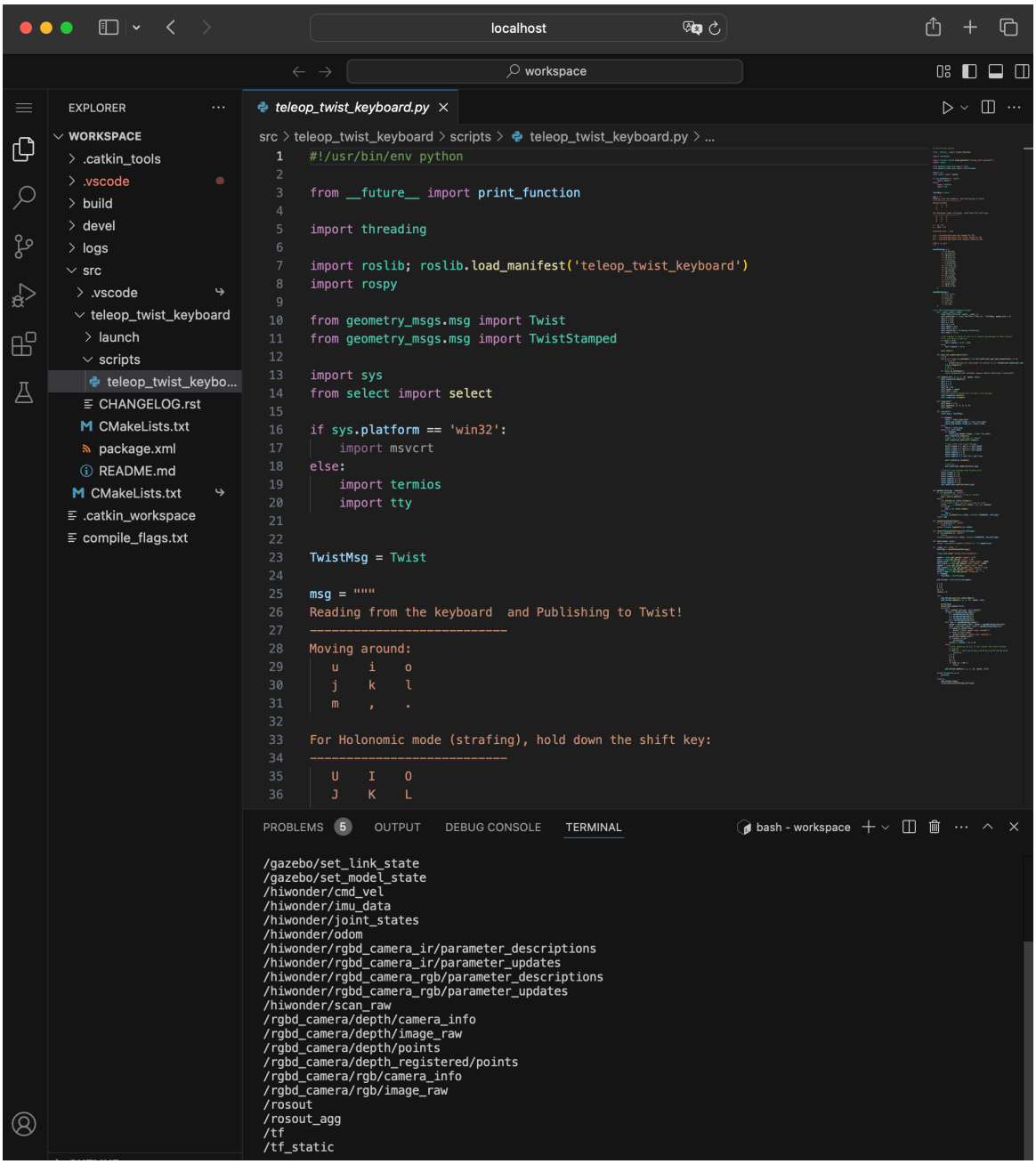
If everything is correctly set up, you should be able to see the Gazebo simulator as below. There is a green-black colour robot in the simulator.



Coding IDEs

We recommend using locally installed VSCode as the IDE to work. However, if you prefer, there are also browser based tools for you to use.

- Jupiter Notebook: It can be accessed through http://localhost:3002/
- Browser based VSCode as shown below, including integrated terminals at the bottom: It can be accessed through http://localhost:3001/?folder=/home/developer/workspace



This is prepared for the Neoscholar Robot Perception, Localization and Navigation course, based on various online resources. We would like to thank the colleagues who shared the resources used in this, and are happy to explicitly acknowledge if needed.